

YIXUAN LI

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Education

Southern University of Science and Technology (SUSTech)

Sep. 2022 – Present

Bachelor of Science in Data Science

Shenzhen, China

- **GPA:** 3.79/4.00
- **Mathematics Courses:** Linear Algebra, Mathematical Analysis, Discrete Mathematics and Its Applications, Foundation of Probability Theory, Mathematical Statistics, Statistical Linear Models, Operational Research and Optimization
- **CS Courses:** Introduction to Computer Programming, Data Structures and Algorithm Analysis, Principles of Database Systems, Distributed Storage and Parallel Computing, Advanced Natural Language Processing, Artificial Intelligence

University of California, San Diego (UCSD)

Mar. 2025 – Jun. 2025

University and Professional Studies (UPS) Program

San Diego, CA, United States

- **GPA:** 4.00/4.00
- **Courses:** Machine Learning Algorithms, AI Search and Reasoning, Topics of Data Science: Trustworthy Machine Learning

Research Interest

- **Trustworthy Machine Learning:** Uncertainty Quantification, Hallucination Detection, Model Interpretability, LLM Alignment, Generative AI
- **AI for Science (Biomedical):** Multimodal Learning, Graph Neural Networks (GNNs), Medical Imaging

Research Experience

SUSTech Machine Learning Lab | Advisor: Prof. Hongxin Wei | SUSTech

Shenzhen, China

Project: Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction

Main Developer

Oct. 2025 – Present

- Built an end-to-end framework for the STP Open Challenge, translating spatial transcriptomics into proteomic distributions on adjacent glioma tissue sections.
- Developed *UNet* and later *GCN/GAT-based graph models*, introducing a patch-based neighbor-sampling strategy to mitigate boundary bias and improve generalization.
- Designed a *multimodal fusion pipeline* integrating ResNet-derived morphological features from H&E Image with RNA embeddings, boosting performance to **0.74** on Top 10 Average Spearman Correlation and ranking **Top 1** on the official leaderboard.
- Generated biologically coherent protein-level spatial maps that preserved tissue architecture and hotspot regions, enabling downstream analyses such as cellular microenvironment profiling.

Project: Benchmarking Uncertainty Quantification Methods in Large Language Models

Independent Researcher

Jun. 2025 – Present

- Built a unified benchmarking pipeline for 10+ *semantic uncertainty quantification methods* (e.g., SE, DSE, SAR) on QA datasets using open-source LLMs, evaluated via AUROC, AUARC, and PRR.
- Standardized data processing and evaluation for reproducible comparison across uncertainty estimation approaches.
- Conducted an extensive literature review on *semantic uncertainty in LLMs*, connecting with hallucination detection and calibration, and published findings in technical blog posts.

Jianqing Shi's Lab | Advisor: Prof. Jianqing Shi | SUSTech

Shenzhen, China

Project: Statistical Analysis of Mental Health Impact among Menopausal Women

Core Collaborator

Sep. 2025 – Nov. 2025

- Analyzed large-scale survey data on menopausal women in Malaysia to examine the relationship between menopausal stages and mental health outcomes.
- Applied statistical methods including *Kruskal-Wallis*, *Fisher-Exact*, *Chi-square*, *ANCOVA*, *CMH*, *Paired t-tests*, and *multiple imputation* for hypothesis testing and subgroup analysis.
- Identified stage-adjusted psychological risk factors and enhanced inference robustness in women's health research.
- Due to data confidentiality, results are restricted to internal reports.

Internship Experience

Machine Learning for Clinical Decision Support in Orthodontics

Jun. 2024 – Aug. 2024

Summer Internship — Mentor: Dr. Ying Shan

Peking University Shenzhen Hospital

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which include cleaning, de-duplication, and standardization.
- Developed *linear and neural network models* based on craniofacial measurement variables to predict patient-specific tooth extraction plans. Achieved promising prediction accuracy and facilitated internal adoption.

Personal Projects & Competitions

Concept Bottleneck in Generative Models: Reproduction Study | Core Member

Apr. 2025 – Jun. 2025

- Reproduced the CVPR paper on Post-hoc Concept Bottleneck frameworks: *Concept Bottleneck Autoencoder (CB-AE)*, *Concept Controller (CC)* using *StyleGAN*, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on *Concept Extra-Interpolation*, *Entanglement*, and *Leakage* to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including *Github code repository*, *presentation slides*, and a public *technical blog*, achieving a course score of **43.4/40 (Top 1)** .

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | Team Leader

Oct. 2024 – Jan. 2025

- Led a team to fine-tune *Gemma-2B* on the *GPT-4 Chinese QA dataset*. Built a complete fine-tuning pipeline covering data preprocessing, LoRA integration, and evaluation. Published notebook can be viewed *here*.
- Integrated *Retrieval-Augmented Generation (RAG)* for data augmentation and *LoRA-based PEFT* for parameter-efficient fine-tuning, resulting in improved dataset diversity, contextual grounding, and instruction-following performance.
- Evaluated model performance using *BLEU*, *ROUGE*, *METEOR*, and *BERTScore metrics*, and performed *cross-lingual robustness analysis* showing stable results across four languages.

Awards

- First-class Scholarship for Outstanding students, SUSTech** (Top 5%) **2023 – 2024, 2024 – 2025**
- Third-class Scholarship for Outstanding students, SUSTech** (Top 15%) **2022 – 2023**
- Second Prize** Awarded in the 2024 China Undergraduate Mathematical Contest in Modeling (CUMCM), in Guangdong division **Oct. 2024**
- Honorable Mention** Awarded in the 2024 Mathematical Contest In Modeling (MCM) **May 2024**
- Excellent Student Award** For outstanding students in SUSTech for three times **2022 – 2025**

Leadership / Extracurricular

Community Involvement

Oct. 2023 – Present

Volunteer

SUSTech

- Participated in online teaching programs, delivering educational lessons to children in rural areas.
- Accumulated over **179 hours** of volunteer service over three years.
- Recognized as an **Outstanding Volunteer** at both the university and School of Science levels.

Media of Shuren College, SUSTech

Sep. 2023 – Aug. 2024

President

SUSTech

- Leading a team with 15 Team members. Created numerous contents/posts for the college’s official media account.
- Received the **Excellent College Student Leader Award** and the **Outstanding College Organizations Award**.

Pickleball Team

Oct. 2023 – Present

Member

SUSTech

- Represented SUSTech in the Second Guangdong Pickleball Competition (Women’s Doubles, **Top 16** finish)
- Participated in the 2023 SUSTech Pickleball Championship, achieving **5th place** in Mixed Doubles.

Skills

- Programming:** Python, R, SQL, Java, JavaScript, HTML, Hadoop
- Development Tools:** PyTorch, Git, Jupyter Notebook, LaTeX, Markdown
- Languages:** English (Fluent, TOEFL 109, GRE 322), Mandarin (Native), Cantonese (Fluent)